



INDIAN INSTITUTE OF TECHNOLOGY, KHARAGPUR

DEPARTMENT OF PHYSICS

END-AUTUMN SEMESTER EXAMINATION 2023-24

Subject: Classical Mechanics (PHPH31207)

Total Marks: 50

Time Duration: 3 Hours

No. of students: 102

Date: 21-11-2023 & Time: 2 PM- 5 PM

Answer all the questions (Start each question on a fresh page)
(If the given information is not sufficient make proper assumptions and mention them clearly)

1. A particle of mass m is constrained to move on the surface of a sphere of radius R . There are no external forces on any kind on the particle.
 - a) What is the Hamiltonian of the system? Is it a constant of motion?
 - b) Prove that the motion of the particle is along the great circle of the sphere (4M+2M)
2. If the Hamiltonian and Lagrangian of a dynamical system is a function of some parameter α , prove that $\left(\frac{\partial H}{\partial \alpha}\right)_{p,q} = \left(\frac{\partial L}{\partial \alpha}\right)_{\dot{q},q}$ (6M)
3. Show that the transformation $p = m\omega q \cot(Q)$ and $P = \frac{m\omega q^2}{2 \sin^2 Q}$ is canonical using standard method (not Poisson Bracket method) and obtain the generator of the transformation. (6M)
4.
 - (a) State and prove Poisson's First theorem.
 - (b) Show that the Poisson Bracket is invariant under canonical transformation
 - (c) Show that the following transformations are canonical using Poisson Brackets
 $Q_1 = \frac{1}{\sqrt{2}}\left(q_1 + \frac{p_2}{m\omega}\right), P_1 = \frac{1}{\sqrt{2}}(p_1 - m\omega q_2), Q_2 = \frac{1}{\sqrt{2}}\left(q_1 - \frac{p_2}{m\omega}\right), P_2 = \frac{1}{\sqrt{2}}(p_1 + m\omega q_2)$
where $m\omega$ is constant. (2M + 6M + 6M)
5. A particle of mass m is thrown horizontally from the top of a building of height h with an initial velocity u . Obtain the equation of motion by using Hamilton-Jacobi method. (10M)
6. Obtain the frequency of the harmonic oscillator by the action-angle variable method. Express the Canonical coordinate and momentum in terms of the action-angle variables. (8 M)

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